MILITARY SPECIFICATION

AMMONIUM PERCHLORATE

This Interim Amendment is approved for use within the US Army Armament, Munitions and Chemical Command, with MIL-A-192B, dated 2 September 1965.

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1.1 Scope. Change "three grades" to "four grades".

1.2 Classification. Add the following grade:

"Grade D - Retained on 74 micron sieve, for use in M864 propellant."

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2.2 Other Publications. Add the following under "CODE OF FEDERAL REGULATIONS":

"Title 49, Transportation, Parts 100-199"

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite: 49CFR 100-199 (latest revision)).

Add the following:

"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E300 - Recommended Practice for Sampling Industrial Chemicals

(Application for copies should be addresssed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)"

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Table I. Add the following to Table I:

"Requirement	Percent by Weight Grade D Min.	Max.	Test Method
Total moisture	-	0.10	4.3.1
Water insolubles	-	0.20	4.3.2
pH of Water Solution	5.0	6.5	4.3.9
Tricalcium Phosphate	0.15	0.22	4.3.12
Chlorate, as NH_4ClO_3	-	0.02	4.3.13
Perchlorate as NH_4ClO_4	98.8	-	4.3.14"

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Table IIa. Add the following table below Table II:

"U.S. Sieve Number	Opening, Microns	Percent Retained Min.	Max.
45	-	0	2.0
200	74	91.0	99.0
325	44	1.0	11.0"

Add the following new paragraph:

"4.2.3.2 Sampling for M864 propellant component. Approximately 500 grams of the composition shall be selected from each batch to be sampled using ASTM Method E300-73 for solids. Samples shall be selected for inspection in accordance with table below. If any sample fails to meet any test requirements the batch represented by the sample shall be rejected. Each requirement test shall be performed in duplicate. No composite samples shall be used.

Number of containers in batch	Number of containers to be sampled
2 - 100	2
101 - 200	5
201 - 300	8
301 - 500	10
501 - 1000	13
1001 - 3000	20
3001 - 10,000	32"

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4.3.10 Granulation. Delete in its entirety and substitute the following:

"4.3.10 Granulation. The granulation (Grades A, B, and C) shall be determined in accordance with MIL-STD-1234, Method 201.1 The granulation (Grade D) shall be determined in accordance with MIL-STD-1234, Method 201.1, with the following changes:

The sample weight shall be 50 g. and results shall be calculated with the following formula:

$$\text{Percent retained on screen} = \frac{A}{W} (100)"$$

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Add the following new paragraphs:

"4.3.12 Tricalcium phosphate (Determination for Grade D.)

4.3.12.1 Reagents.

a. Ammonium Molybdate Solution, 10% aqueous Perchloric acid

b. Ammonium Phosphate (dibasic) 0.213 g/250 ml

c. Ammonium Vanadate Solution, 0.25% aqueous

4.3.12.2 Apparatus. Bausch & Lamb "Spectronic" 710 or equivalent.

4.3.12.3 Preparation of test solutions.

a. Blank Solution: In a clean 100 ml volumetric flask, place 25 ml of distilled water and 17 ml of concentrated perchloric acid. Cool to room temperature and add 10 ml of ammonium vanadate solution (0.25%). Dilute to about 75 ml with water and add 10 ml of ammonium molybdate solution (10%). Dilute to mark volume with water, mix well, and allow to stand for 30 minutes for maximum color development.

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b. Standard Solutions: In two clean 100 ml volumetric flasks, place respectively, 1.0 and 2.0 ml of ammonium phosphate standard solution. In order, treat each flask as the blank was treated. The concentrations of the standard solutions equate to 0.10 and 0.20% TCP, respectively.

c. Preparation of Sample Solutions: Weigh a 25 gm sample to the nearest 0.1 mg and transfer it to a 250 ml volumetric flask. Add about 50 ml of water and mix well. Carefully add 25 ml of concentrated perchloric acid to the flask, swirling constantly, and cool to room temperature. Dilute to volume and mix well. Pipet a 10 ml aliquot into a 100 ml volumetric flask. Add 20 ml of water. Add 16 ml of concentrated perchloric acid. Add 10 ml of ammonium vanadate (0.25%), dilute to about 75 ml with water, swirling constantly, and cool to room temperature. Add 10 ml of ammonium molybdate (10%), dilute to volume, and mix well. Allow the solution to stand for at least 30 minutes at room temperature.

4.3.12.4 Procedure.

a. Set wavelength at 470 nanometers (nm) and introduce the blank solution. Turn selector to concentration mode and set scale readout to zero. Introduce the 0.20% TCP standard solution and, by use of the concentration adjustment control, set the readout to 0.20.

b. For linearity verification, introduce the 0.10% TCP standard solution. The display readout should show 0.10. If linearity is not achieved, the solutions should be reintroduced in order, with appropriate control adjustment and, if this fails, new blank and standard solutions prepared. Introduce sample solutions prepared in the exact manner described, and read the sample concentration, in percent TCP, directly from the instrument display readout. Additional instructions may be obtained from the instrument operator's manual.

4.3.13 Chlorate (Determination for Grade D.

4.3.13.1 Reagents.

a. 1:3 Hydrochloric Acid

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b. Standard 0.06 g/l Potassium Chlorate Solution

c. O-Tolidine Solution

4.3.13.2 Apparatus. Bausch & Lomb "Spectronic" 710 or equivalent.

4.3.13.3 Preparation of test solutions.

a. Blank Solution: Add 80 ml of 1:3 hydrochloric acid solution to a clean 100 ml volumetric flask and heat to 40°C for about 30 minutes.

b. Standard Solutions: In a clean 100 ml volumetric flask, add 1.0 ml of the standard 0.06 g/l potassium chlorate solution and 80 ml of the 1:3 hydrochloric acid. To another clean 100 ml volumetric flask, add 2.0 ml of the potassium chlorate solution and 80 ml of 1:3 hydrochlorate solution and 80 ml of 1:3 hydrochloric acid. Heat both flasks to 40°C for 30 minutes. The concentrations of the standard solutions equate to 0.50 and 1.0 micrograms of ammonium chlorate per milliliter, respectively, or concentrations of 0.005 and 0.01 percents ammonium chlorate when applied to a 1.000 g sample.

c. Preparation of Sample Solutions: Weigh 1 gm of sample, accurate to ± 0.1 mg, and place in a clean 100 ml volumetric flask. Add 80 ml of 1:3 hydrochloric acid solution and heat at 40°C for 30 minutes. Develop color in blank, standards, and sample solutions by cooling to room temperature, adding 1 ml of O-Tolidine solution and diluting to mark volume with water. After mixing, the solutions are allowed to stand for 5 minutes to complete color development.

4.3.13.4 Procedure.

a. Set wavelength at 490 nanometers (nm) and introduce the blank solution. Turn selector to concentration mode and set scale readout to zero. Introduce the high concentration standard solution, 0.01% NH_4ClO_3, and, by use of the concentration adjustment control, set the readout to 0.01.

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b. For linearity verification, introduce the 0.05% standard solution, and the readout should show 0.05. If linearity is not achieved, the solutions should be reintroduced in order, with appropriate control adjustment, and, if this fails, new blank and standard solutions prepared. Introduce sample solutions,prepared in the exact manner described, and read the sample concentration in percent ammonium chlorate (NH_4ClO_3) directly from the instrument display readout. Additional instructions may be obtained from the instrument operator's manual.

4.3.14 Perchlorate. (Determination for Grade D.)

a. Accurately weigh 5.0 gm of dried sample and dissolve in 40 ml of water in a glass-stoppered flask.

b. Mix 20 ml of formaldehyde with 20 ml of water and add to flask.

c. Add 3 drops of phenolphthalein and neutralize with 1N NaOH.

d. Mix, allow to stand 30 minutes, then add 1N NaOH to a pink color which persists for 5 minutes.

Calculation:

$$\% NH_4ClO_4 = \frac{11.75\ (A)\ (N)}{W}$$

Where:

A = ml of NaOH used for titration
N = normality of NaOH
W = weight of sample"

5. PACKAGING: Delete paragrahs 5.1 and 5.2 in their entirey and substitute the following:

"5.1 Packaging and packing.

5.1.1 Level A. Ammonium perchlorate shall be packed in 250 pound quantities in 30 gallon capacity drums conforming to CFR, TITLE 49, Parts 100-199, DOT specification 6B. The drums shall be cleaned, phosphate coated, and painted in conformance with the requirements of PPP-D-729. Paint shall conform to TT-E-485, color

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No. 24064 of Federal Standard 595. The drums shall be provided with bag liners fabricated from 2 mil (min) polyethlene conforming to L-P-378. The bag liners shall have flat or square bottoms with heat sealed seams that meet the requirements of MIL-P-116. The bag shall be closed either by pressure sensitive tape conforming to L-T-100 or by heat seal (Caution: Prevent contact of the ammonium perchlorate with the heat sealer). Two 8-unit bags of silica gel, conforming to MIL-D-3464, shall be inserted in each drum.

5.1.2 Level B. Same as Level A.

5.1.3 Level C. The ammonium perchlorate shall be packed in uniform quantities in conformance with manufacture's commercial practice in accordance with CFR, TITLE 49, Parts 100-199, Paragraph 173.239a. Containers for like class material shall be of uniform size, shape, and material. Containers shall be provided with a polyethylene liner a specified in 5.1.1.

5.2 MARKING

5.2.1 Level A. In addition to any special marking required by the contract or order, all shipping containers packed in accordance with level A shall be marked in accordance with MIL-STD-129 and DOT regulations.

5.2.2 Level B. Same as Level A.

5.2.3 Level C. Same as Level A."

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Add the following paragraph:

"6.1.3 Grade D. This ammonium perchlorate is intended for use in the M864 propellant."

Custodian:
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Preparing activity
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(Project 6810-A009)

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